Robots and Jobs: Evidence from US Labor Markets

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Abstract / Résumé / 要旨

This paper studies the effects of industrial robots on US labor markets between 1990 and 2007. Using a model in which robots compete against human workers in the production of different tasks, the authors show that robots may reduce employment and wages. They estimate that one more robot per thousand workers reduces the employment-to-population ratio by about 0.2 percentage points and wages by about 0.42%. The employment effects are most pronounced in manufacturing, and in particular, in routine manual, blue-collar occupations that do not require significant cognitive flexibility. Interestingly, the authors find negative effects on essentially all occupations, with the exception of managers. The findings suggest that the displacement effect of automation technologies is a significant force shaping labor market outcomes in the United States.

Keywords / Mots-clés / キーワード

automation; robots; employment; wages; labor markets; technological change; United States

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